



US 20250277490A1

(19) **United States**(12) **Patent Application Publication**
KWON et al.(10) **Pub. No.: US 2025/0277490 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **ROTARY COMPRESSOR HAVING FLAT MUFFLER**(71) Applicant: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)(72) Inventors: **Munseong KWON**, Suwon-si (KR);
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Jongwon CHOI, Suwon-si (KR)(73) Assignee: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)(21) Appl. No.: **19/210,510**(22) Filed: **May 16, 2025****Related U.S. Application Data**

(63) Continuation of application No. PCT/KR2023/016758, filed on Oct. 26, 2023.

(30) **Foreign Application Priority Data**

Nov. 18, 2022 (KR) 10-2022-0155777

Publication Classification(51) **Int. Cl.**
F04C 29/06 (2006.01)(52) **U.S. Cl.**
CPC **F04C 29/065** (2013.01); **F04C 2240/30** (2013.01)(57) **ABSTRACT**

A rotary compressor includes: a casing; a motor disposed inside the casing; a compression part disposed below the motor; a lower flange disposed below the compression part; a muffler, having a portion that is flat, disposed on a lower surface of the lower flange; and a sealing plate disposed between the muffler and the lower flange.

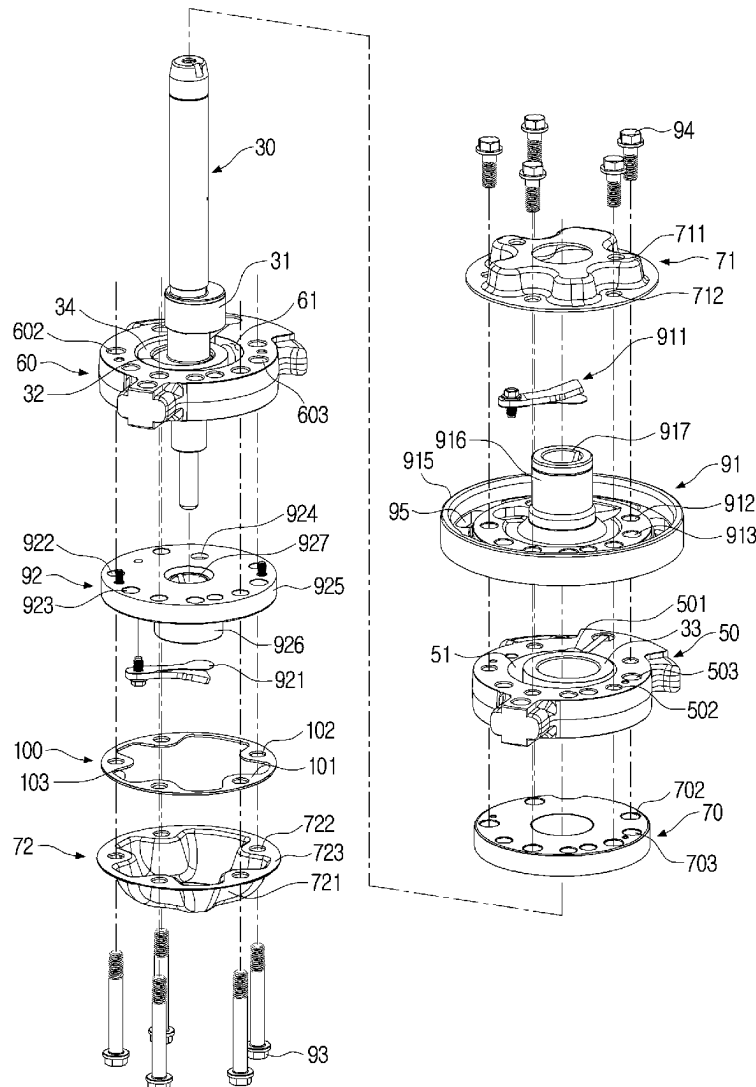


FIG. 1

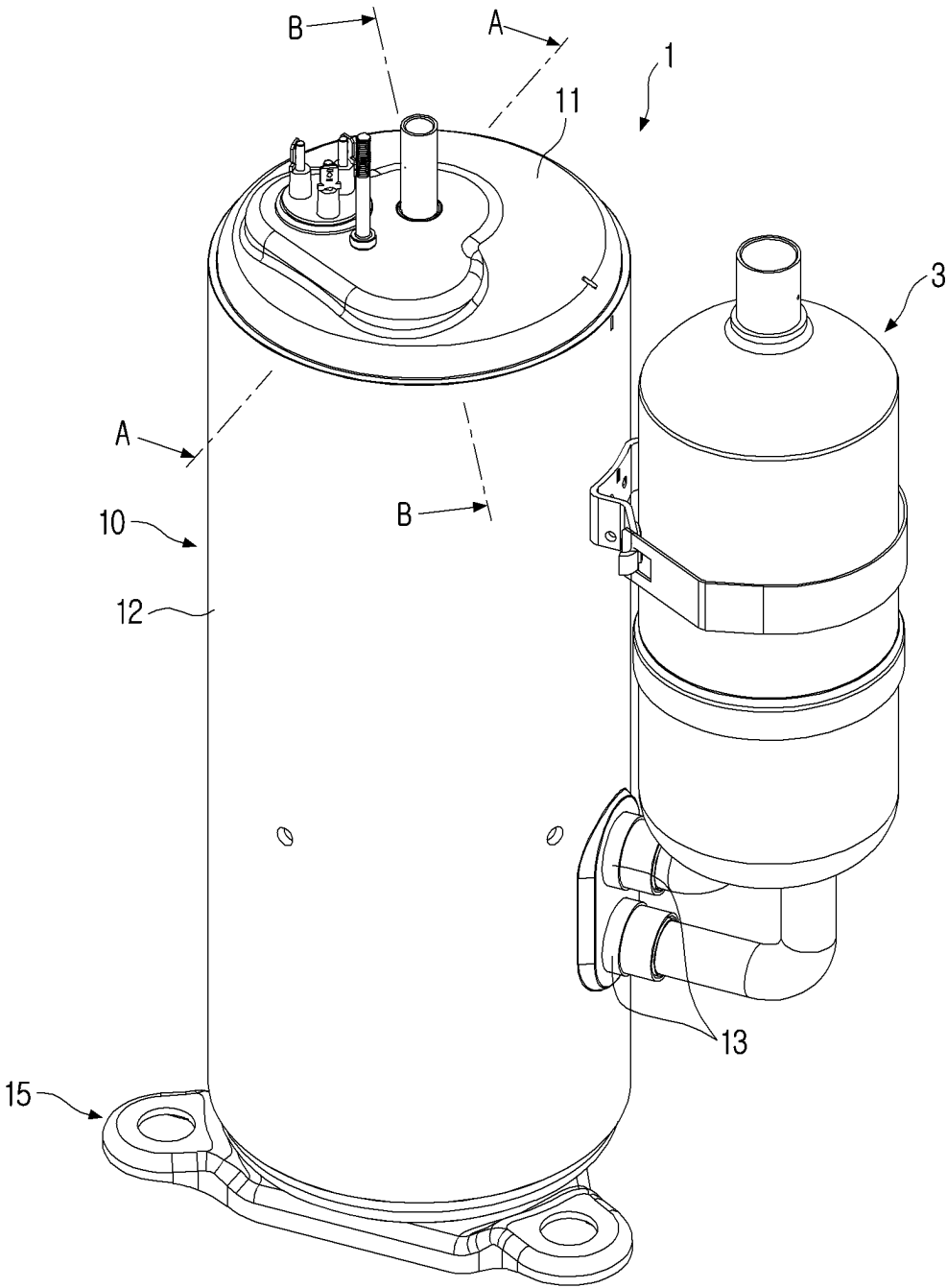


FIG. 2

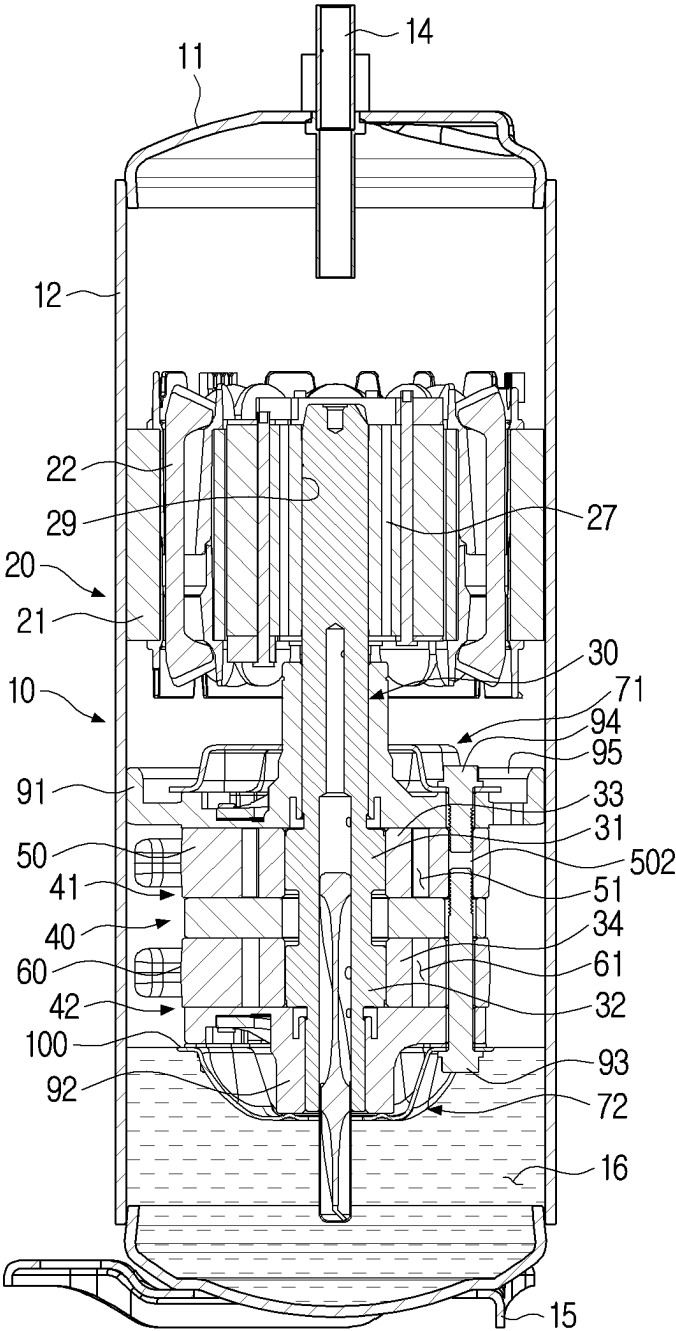


FIG. 3

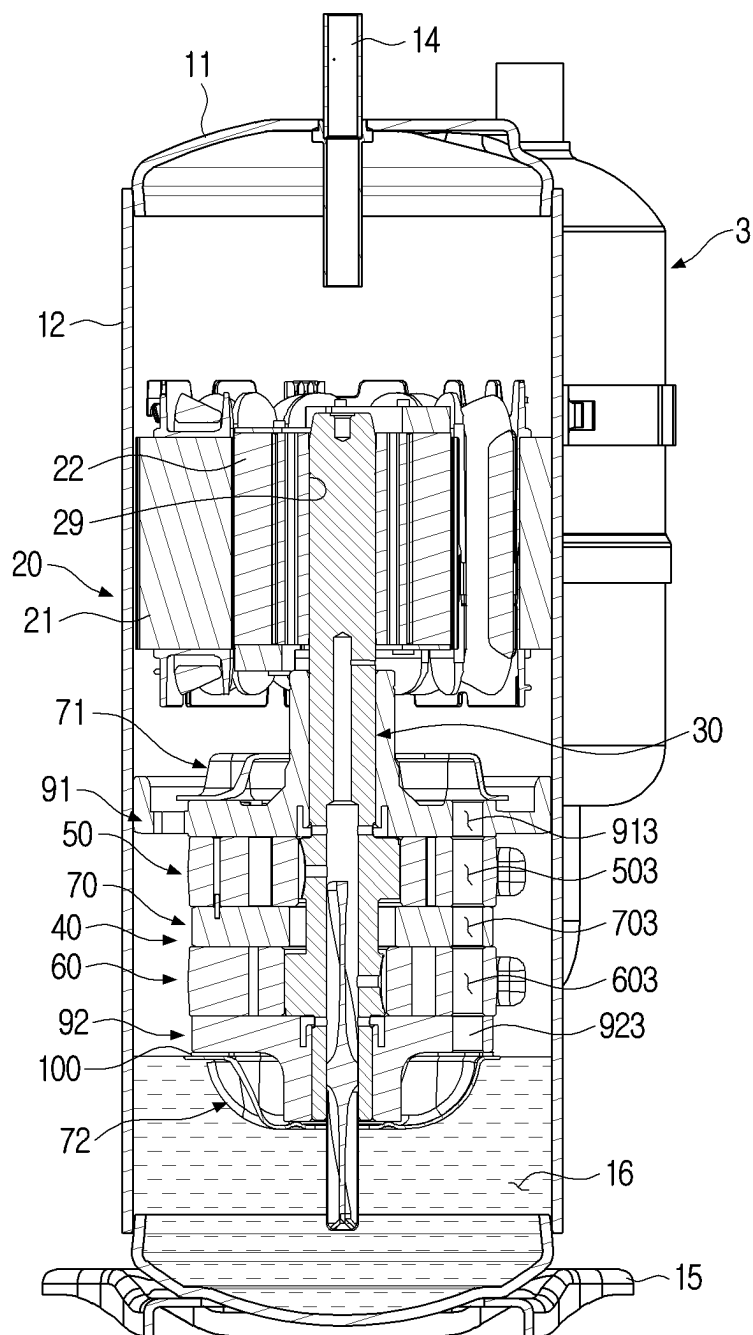


FIG. 4

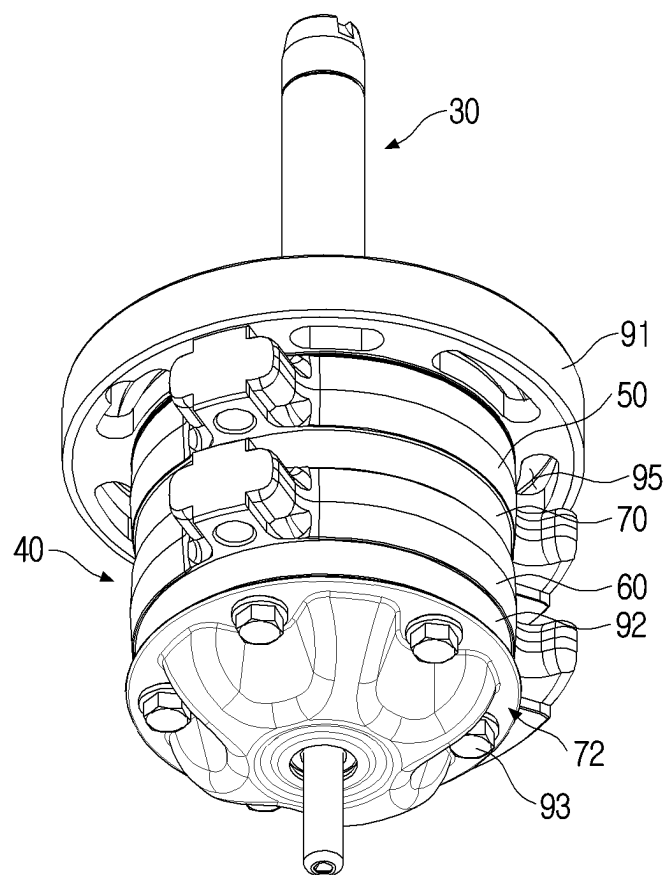


FIG. 5

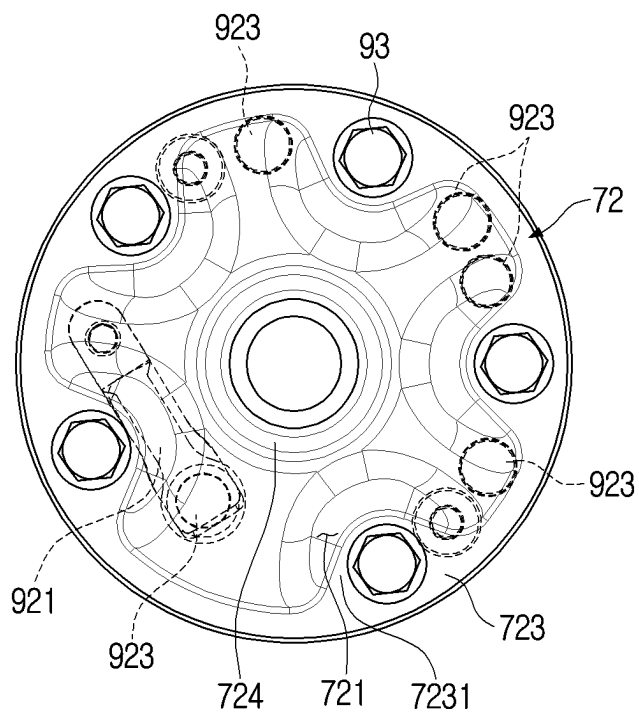


FIG. 6

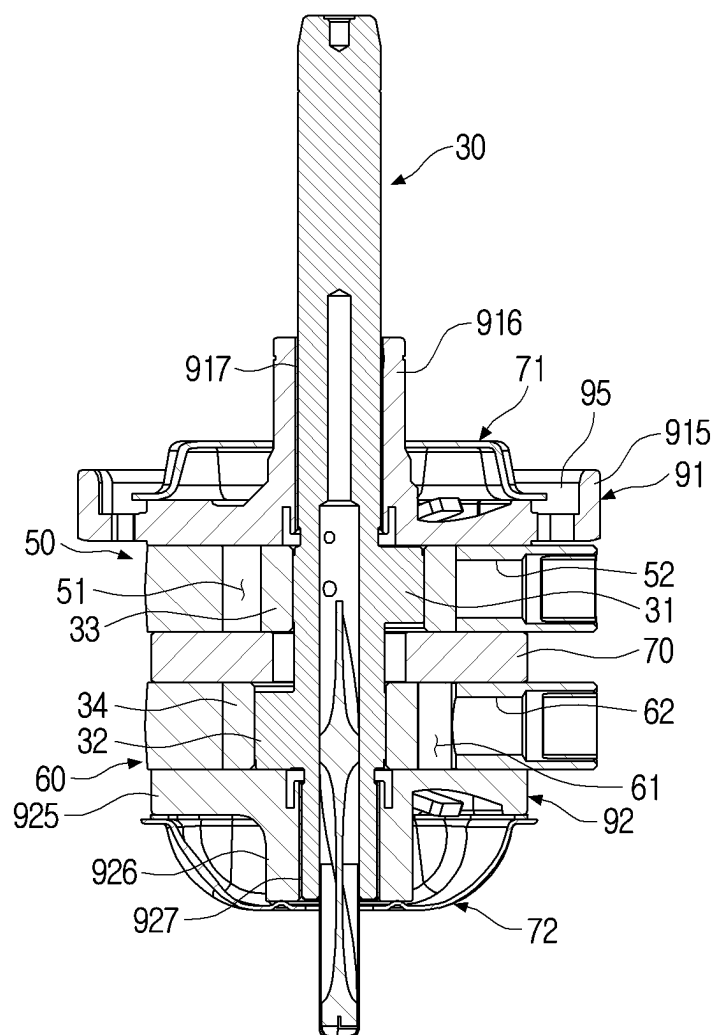


FIG. 7

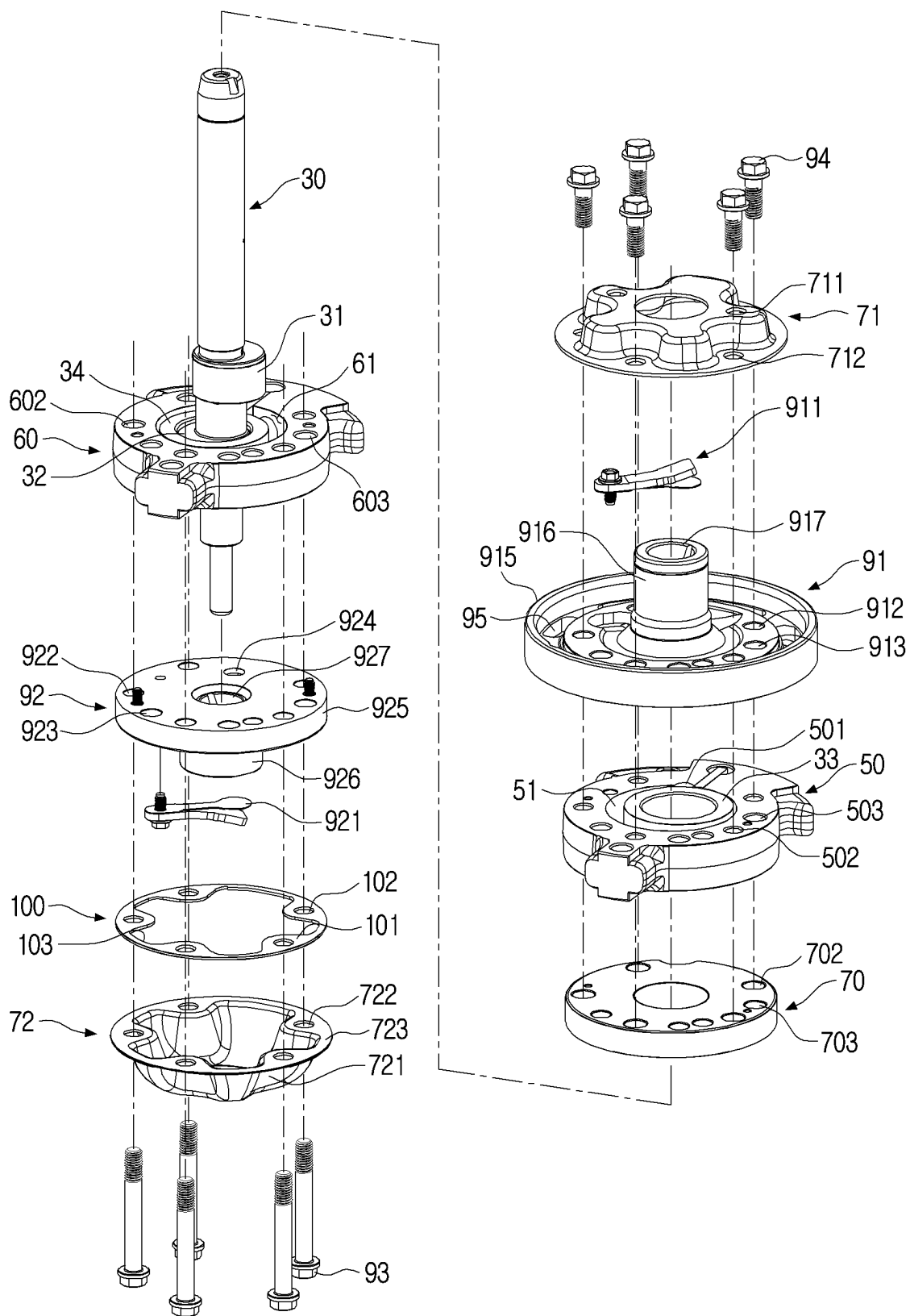


FIG. 8

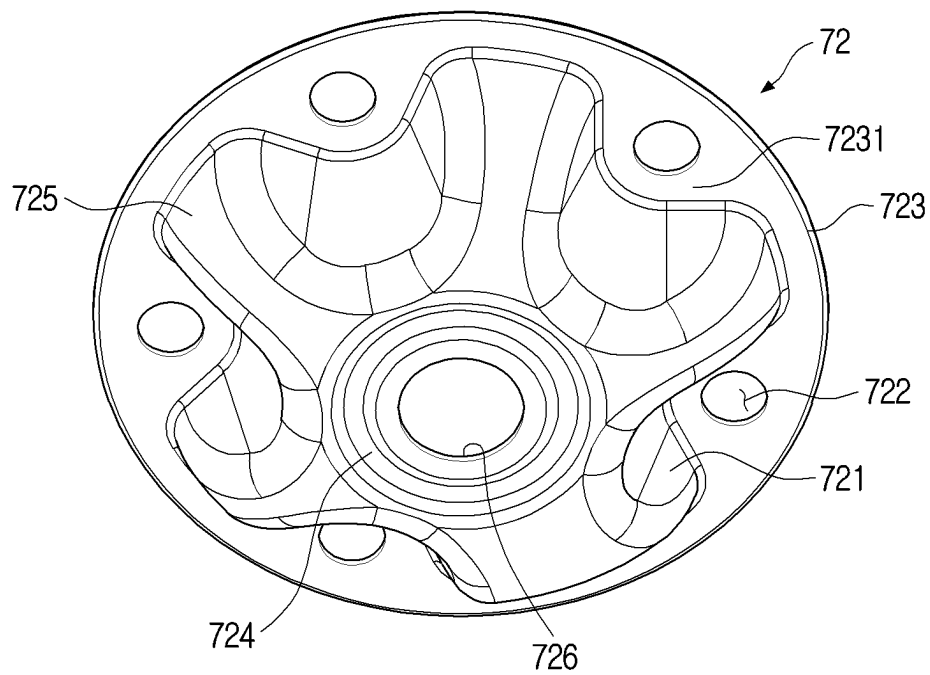


FIG. 9

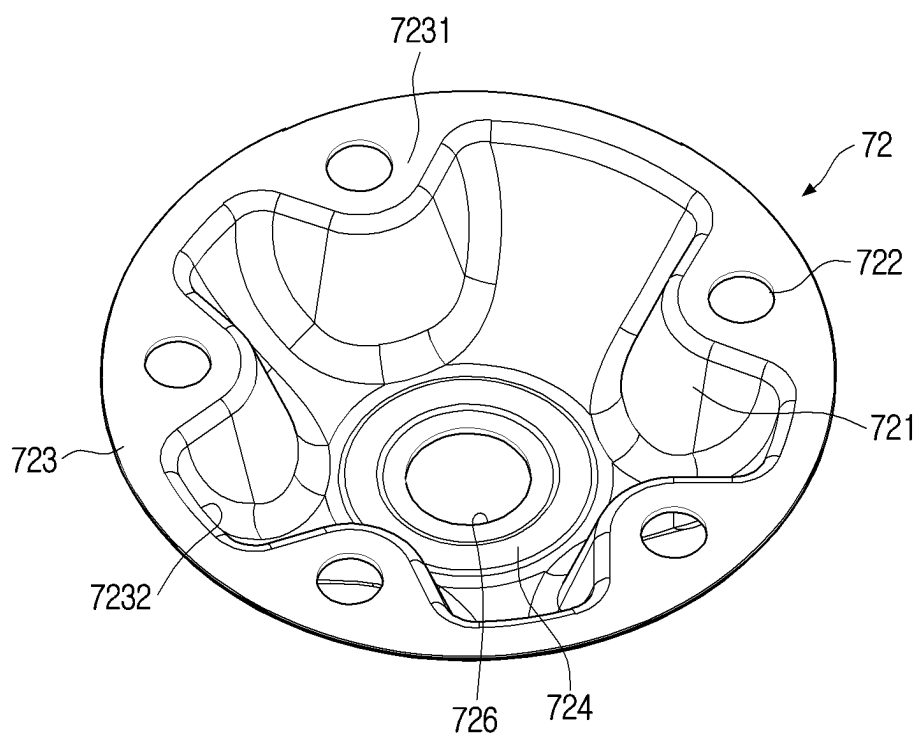


FIG. 10

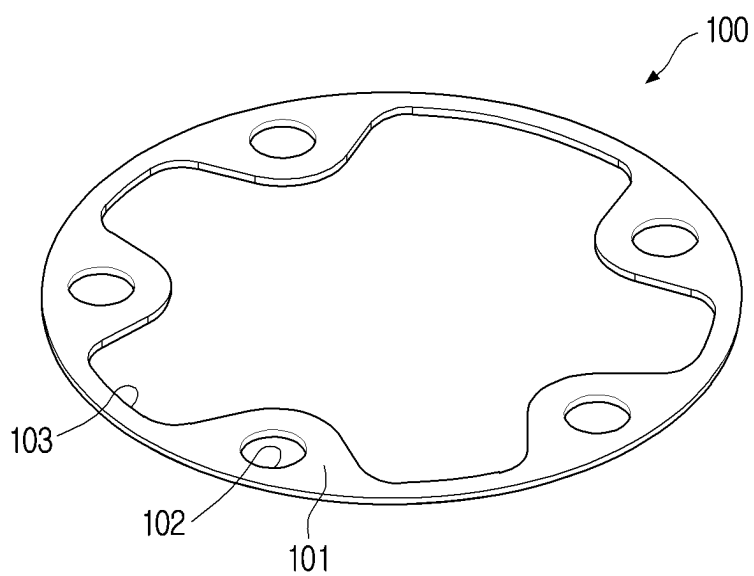


FIG. 11

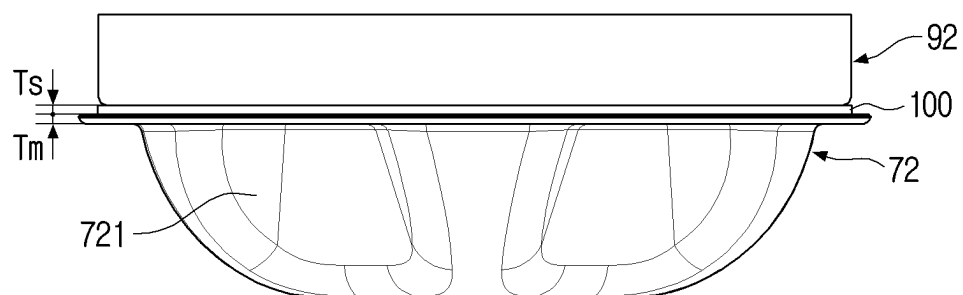


FIG. 12

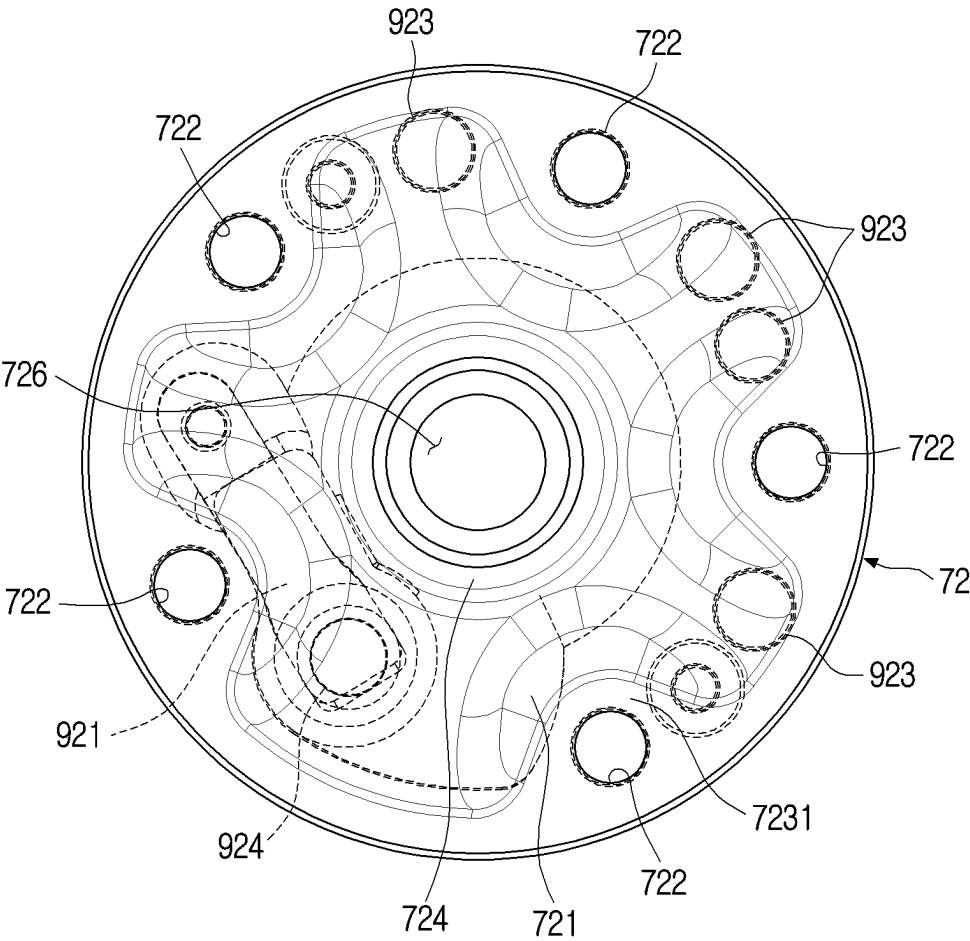


FIG. 13

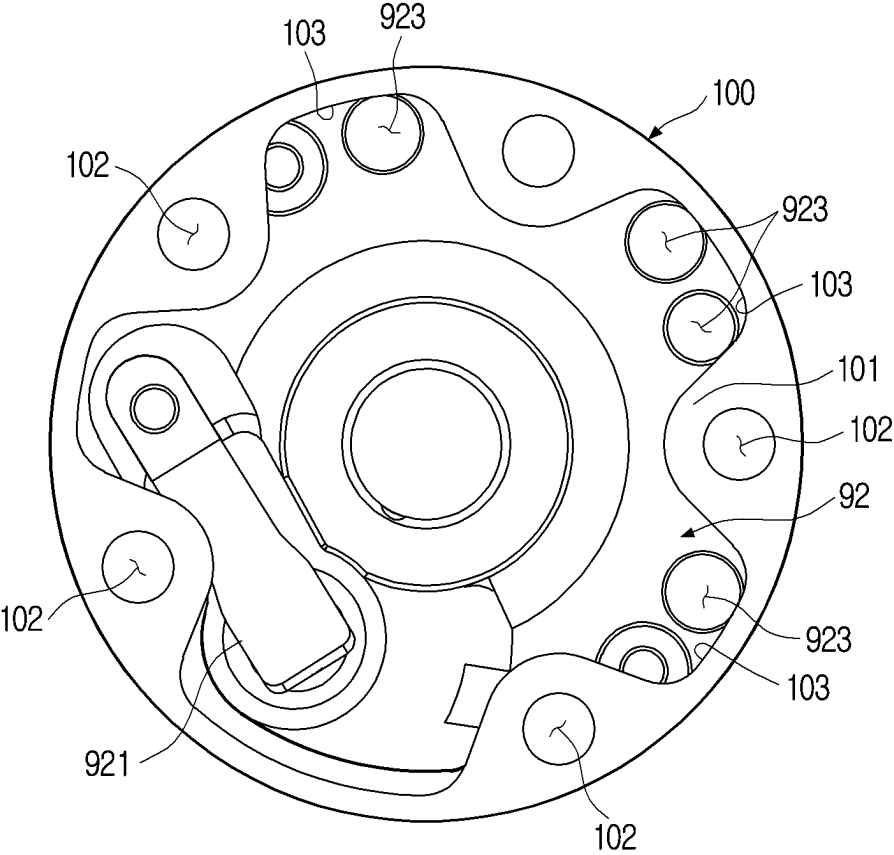


FIG. 14

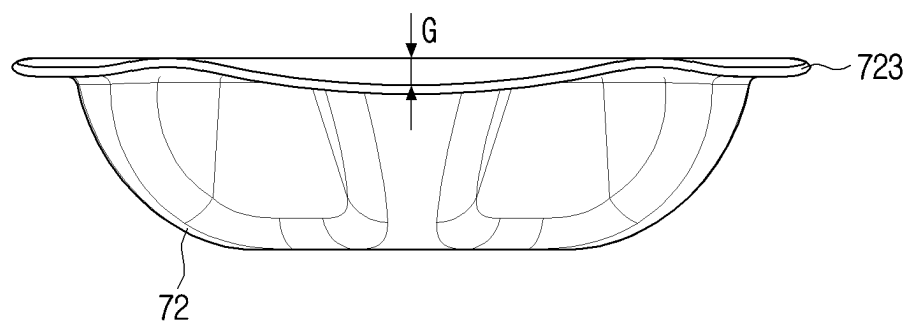


FIG. 15

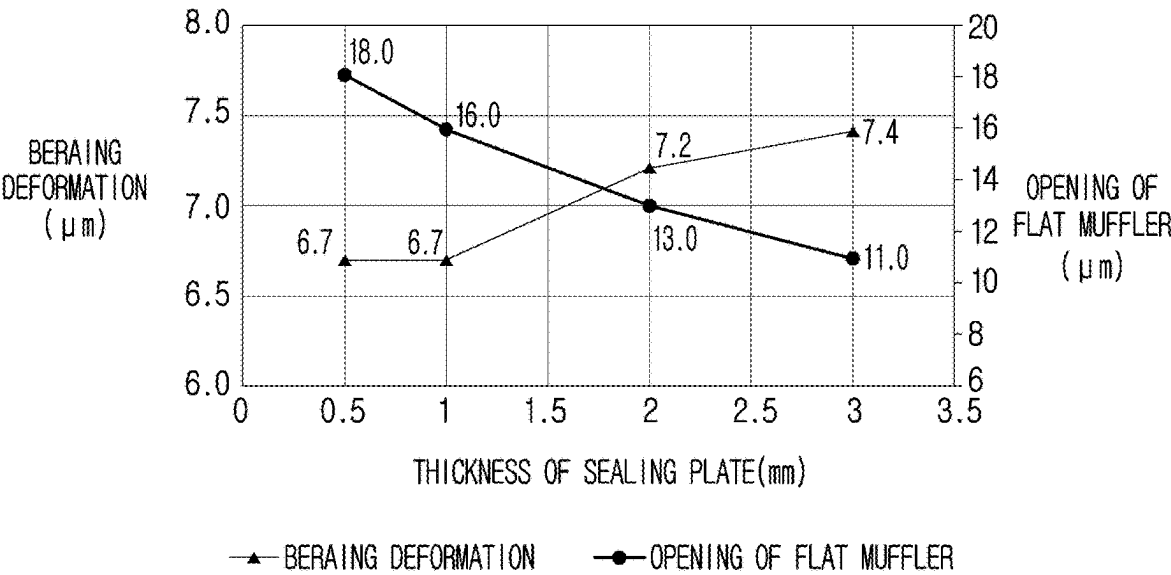
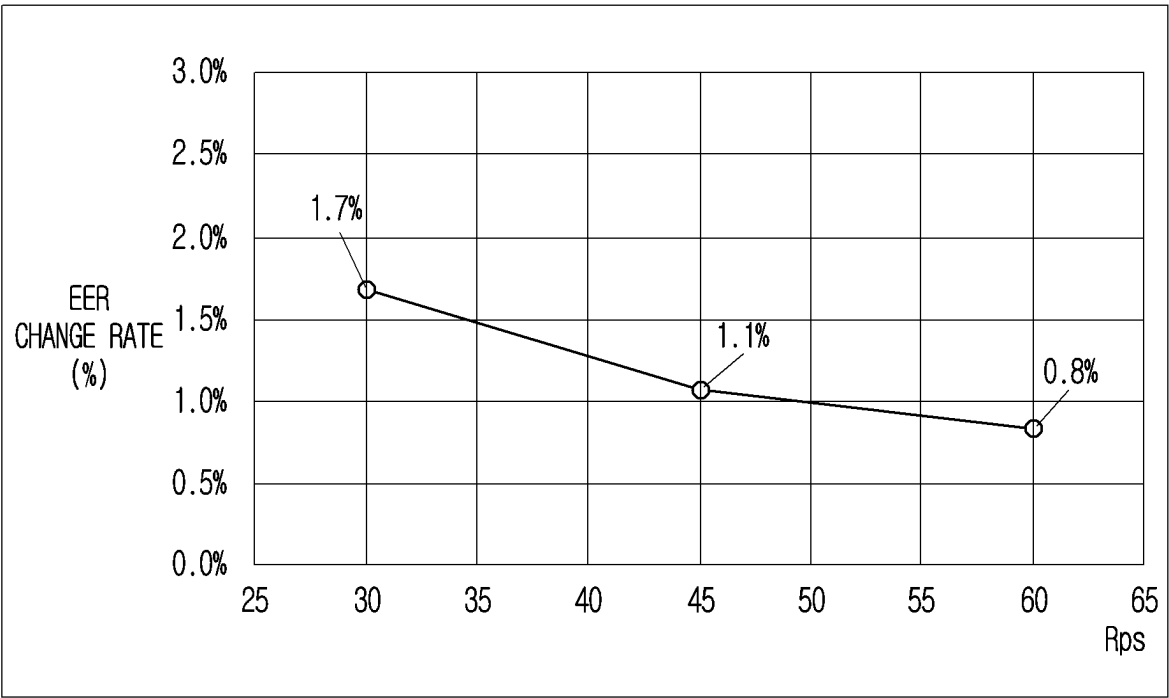


FIG. 16



ROTARY COMPRESSOR HAVING FLAT MUFFLER

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation application is a continuation application, under 35 U.S.C. § 111(a), of international application No. PCT/KR2023/016758, filed Oct. 26, 2023, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0155777, filed Nov. 18, 2022, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a rotary compressor, and more particularly, to a rotary compressor having a flat muffler.

BACKGROUND ART

[0003] A compressor is a mechanical device that compresses gas to increase pressure, and may be classified into reciprocating compressors and rotating compressors based on their operating principles.

[0004] The reciprocating compressor may include a reciprocating compressor that converts the rotational motion of a motor into the linear reciprocating motion of a piston using a crank shaft and a connecting rod to suck and compress gas.

[0005] The rotating compressor may include a rotary compressor and a scroll compressor.

[0006] The rotary compressor may be configured so that refrigerant is sucked and compressed by a roller rotating inside a cylinder of a compression part by the rotational motion of a motor.

[0007] The scroll compressor may be configured so that refrigerant is sucked and compressed by an orbiting scroll rotating in a certain direction relative to a fixed scroll by the rotational motion of a motor.

[0008] In the rotary compressor, the compressed refrigerant may be discharged inside a casing and discharged to the outside through a refrigerant discharge pipe together with oil.

DISCLOSURE OF INVENTION

Technical Solution

[0009] According to an aspect of the disclosure, a rotary compressor may include: a casing; a motor disposed inside the casing; a compression part disposed below the motor; a lower flange disposed below the compression part; a muffler, having a portion that is flat, to be disposed on a lower surface of the lower flange; and a sealing plate disposed between the flat muffler 72 and the lower flange.

[0010] The portion that is flat is an upper end of the flat muffler may not protrude above the lower surface of the lower flange.

[0011] The muffler may be formed in a dome shape. The muffler may include a plurality of concave portions formed at regular intervals along a circumferential direction of the muffler on a side surface of the muffler. The sealing plate may be formed in a ring shape. An inner circumferential surface of the sealing plate may be formed concavely and convexly corresponding to the muffler.

[0012] The lower flange may include a plurality of lower refrigerant holes and a plurality of first bolt holes formed along an edge of the lower flange. The muffler may include a plurality of second bolt holes formed corresponding to the plurality of first bolt holes of the lower flange at positions corresponding to the plurality of concave portions along an edge of the muffler.

[0013] The sealing plate may include a plurality of bolt seats formed to protrude toward a center of the sealing plate from an inner circumferential surface of the sealing plate and having a plurality of third bolt holes corresponding to the plurality of second bolt holes of the muffler.

[0014] The sealing plate may be formed so as not to cover the plurality of lower refrigerant holes of the lower flange.

[0015] A minimum distance between the inner circumferential surface of the sealing plate 100 and a plurality of lower refrigerant holes of the lower flange may be zero (0).

[0016] A thickness of the sealing plate may have a following relationship with a thickness of the muffler: $0.5 \leq \text{thickness } T_s \text{ of the sealing plate } 100 / \text{thickness } T_m \text{ of the muffler } 72 \leq 2.0$.

[0017] Oil may be received in a lower portion of the casing, and the muffler may be immersed in the oil.

[0018] The sealing plate may be formed of one of heat-resistant resin, steel, copper, and a material in which at least two of the heat-resistant resin, the steel, and the copper are laminated.

[0019] The lower flange may include: a flange part formed in a disk shape; a boss extending vertically from the flange part and including a through hole; and a bearing disposed in the through hole of the boss. The muffler may include: a fixing plate formed in a ring shape and fixed to the flange part of the lower flange; a caulking portion fixed to one end of the boss; and a muffler portion provided between the fixing plate and the caulking portion.

[0020] The muffler portion may include a plurality of concave portions formed at regular intervals in a circumferential direction of the muffler.

[0021] The fixing plate may include a plurality of protrusions corresponding to the plurality of concave portions of the muffler portion.

[0022] The flange part of the lower flange may include a plurality of lower refrigerant holes and a plurality of first bolt holes. The plurality of protrusions of the fixing plate may include a plurality of second bolt holes corresponding to the first bolt holes.

[0023] An inner circumferential surface of the sealing plate may include a plurality of bolt seats formed to correspond to the plurality of protrusions of the fixing plate and a plurality of refrigerant grooves formed so as not to cover the plurality of lower refrigerant holes of the flange part.

BRIEF DESCRIPTION OF DRAWINGS

[0024] These and/or other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0025] FIG. 1 is a perspective view illustrating a rotary compressor according to an embodiment of the disclosure.

[0026] FIG. 2 is a cross-sectional view illustrating the rotary compressor of FIG. 1 taken along line A-A according to an embodiment of the disclosure.

[0027] FIG. 3 is a cross-sectional view illustrating the rotary compressor of FIG. 1 taken along line B-B according to an embodiment of the disclosure.

[0028] FIG. 4 is a perspective view illustrating a compression part of a rotary compressor according to an embodiment of the disclosure.

[0029] FIG. 5 is a bottom view illustrating the compression part of the rotary compressor of FIG. 4 according to an embodiment of the disclosure.

[0030] FIG. 6 is a cross-sectional view illustrating the compression part of the rotary compressor of FIG. 4 according to an embodiment of the disclosure.

[0031] FIG. 7 is an exploded perspective view illustrating the compression part of the rotary compressor of FIG. 4 according to an embodiment of the disclosure.

[0032] FIG. 8 is a perspective view illustrating a flat muffler used in a rotary compressor according to an embodiment of the disclosure.

[0033] FIG. 9 is a perspective view illustrating the flat muffler of FIG. 8 as seen from above according to an embodiment of the disclosure.

[0034] FIG. 10 is a perspective view illustrating a sealing plate used in a rotary compressor according to an embodiment of the disclosure.

[0035] FIG. 11 is a view illustrating a state in which a sealing plate and a flat muffler are disposed in a lower flange according to an embodiment of the disclosure.

[0036] FIG. 12 is a bottom view of FIG. 11 according to an embodiment of the disclosure.

[0037] FIG. 13 is a bottom view illustrating a state in which the flat muffler is removed from FIG. 11 according to an embodiment of the disclosure.

[0038] FIG. 14 is a view illustrating a state in which a portion of a flat muffler is opened by compressed refrigerant according to an embodiment of the disclosure.

[0039] FIG. 15 is a graph illustrating a change in deformation of a bearing and opening of a flat muffler according to a thickness of a sealing plate according to an embodiment of the disclosure.

[0040] FIG. 16 is a graph illustrating a change rate in energy efficiency of a rotary compressor according to an embodiment of the disclosure relative to the energy efficiency of a rotary compressor according to the prior art using a skirt muffler.

BEST MODE FOR CARRYING OUT THE INVENTION

[0041] Since the embodiments of the disclosure can apply various transformations and have various embodiments, specific embodiments will be illustrated in the drawings and described in detail in the detailed description. However, this is not intended to limit the scope to the specific embodiments, and should be understood to include various modifications, equivalents, and/or alternatives of the embodiment of the disclosure. In connection with the description of the drawings, like reference numerals may be used for like elements.

[0042] In describing the disclosure, when it is determined that a detailed description of a related known function or configuration may unnecessarily obscure the gist of the disclosure, a detailed description thereof will be omitted.

[0043] In addition, the following embodiments may be modified in many different forms, and the scope of the technical idea of the disclosure is not limited to the follow-

ing embodiments. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the spirit of the disclosure to those skilled in the art.

[0044] Terms used in this disclosure are only used to describe specific embodiments, and are not intended to limit the scope of rights. Singular expressions include plural expressions unless the context clearly dictates otherwise.

[0045] In this disclosure, expressions such as “has,” “can have,” “includes,” or “can include” indicate the existence of a corresponding feature (e.g., numerical value, function, operation, or component such as a part) and do not preclude the existence of additional features.

[0046] In this disclosure, expressions such as “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may include all possible combinations of the items listed together. For example, “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may refer to all cases (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

[0047] Expressions such as “first,” “second,” “primary,” or “secondary,” as used in this disclosure may modify various components regardless of order and/or importance, are used only to distinguish one component from other components, and do not limit the corresponding components.

[0048] Further, terms such as ‘leading end’, ‘rear end’, ‘upper side’, ‘lower side’, ‘top end’, ‘bottom end’, etc. used in the disclosure are defined with reference to the drawings. However, the shape and position of each component are not limited by these terms.

[0049] Hereinafter, embodiments of a rotary compressor according to the disclosure will be described in detail with reference to the attached drawings.

[0050] FIG. 1 is a perspective view illustrating a rotary compressor 1 according to an embodiment of the disclosure.

[0051] Referring to FIG. 1, a rotary compressor 1 according to an embodiment of the disclosure may include a casing 10.

[0052] The casing 10 may form an outer appearance of the rotary compressor 1. The casing 10 may be configured as a sealed container. The casing 10 may include a refrigerant inlet 13 through which refrigerant is introduced and a refrigerant discharge pipe 14 through which refrigerant is discharged.

[0053] The rotary compressor 1 may form a refrigeration cycle together with a condenser, an expansion valve, and an evaporator. In this case, the refrigerant inlet 13 may be connected to the evaporator, and the refrigerant discharge pipe 14 may be connected to the condenser.

[0054] An accumulator 3 may be disposed on one side of the rotary compressor 1. In this case, the refrigerant inlet 13 may be connected to the accumulator 3. The inlet pipe of the accumulator 3 may be connected to the evaporator. Therefore, the refrigerant discharged from the evaporator may be introduced into the rotary compressor 1 through the accumulator 3.

[0055] The casing 10 may include an upper casing 11 and a lower casing 12. The upper casing 11 may be connected to the upper end of the lower casing 12 to form the casing 10.

[0056] The joint of the upper casing 11 and the lower casing 12 may be sealed.

[0057] The refrigerant discharge pipe 14 may be provided in the upper casing 11. The refrigerant discharge pipe 14 may be provided at the upper end of the upper casing 11.

[0058] The refrigerant inlet 13 may be provided at the lower casing 12. The refrigerant inlet 13 may be connected to a compression part 40 disposed inside the lower casing 12. A low-temperature/low-pressure refrigerant may be introduced into the refrigerant inlet 13. Therefore, the refrigerant may be introduced into the compression part 40 through the refrigerant inlet 13.

[0059] The accumulator 3 may be disposed in the lower casing 12. In this case, the refrigerant inlet 13 may be connected to the discharge pipe of the accumulator 3.

[0060] A base 15 supporting the casing 10 may be provided at the bottom of the lower casing 12. The rotary compressor 1 may be disposed vertically to the support surface by the base 15.

[0061] FIG. 2 is a cross-sectional view illustrating the rotary compressor 1 of FIG. 1 taken along line A-A. FIG. 3 is a cross-sectional view illustrating the rotary compressor 1 of FIG. 1 taken along line B-B.

[0062] Referring to FIGS. 2 and 3, the rotary compressor 1 according to an embodiment of the disclosure may include the casing 10, a motor 20, and the compression part 40.

[0063] The casing 10 may form the outer appearance of the rotary compressor 1, and be a cylindrical sealed container. The casing 10 may include the lower casing 12 provided with the refrigerant inlet 13 and the upper casing 11 provided with the refrigerant discharge pipe 14.

[0064] The casing 10 may be configured by connecting the upper casing 11 and the lower casing 12, and the interior of the casing 10 except for the refrigerant inlet 13 and the refrigerant discharge pipe 14 may be sealed. In other words, the refrigerant may be introduced into the interior of the casing 10 or discharged to the outside from the casing 10 only through the refrigerant inlet 13 and the refrigerant discharge pipe 14.

[0065] The internal space of the casing 10 may accommodate high-pressure refrigerant discharged from the compression part 40, and the high-pressure refrigerant may be discharged to the outside through the refrigerant discharge pipe 14.

[0066] An oil reservoir 16 for receiving oil may be provided at the lower portion of the casing 10.

[0067] The accumulator 3 may be disposed on the outer surface of the casing 10. At this time, the refrigerant inlet 13 may be connected to the discharge pipe of the accumulator 3.

[0068] The motor 20 may be arranged in the upper side inside the casing 10. The motor 20 may include a stator 21 and a rotor 22.

[0069] The stator 21 of the motor 20 may be fixed to the inner circumferential surface of the casing 10. A plurality of oil return passages may be provided between the outer circumferential surface of the stator 21 and the inner circumferential surface of the casing 10. The plurality of oil return passages may be formed at regular intervals along the outer circumferential surface of the stator 21.

[0070] Oil from the upper side of the motor 20 may move to the lower side of the motor 20 through the plurality of oil return passages provided between the stator 21 and the casing 10. Oil that has moved to the lower side of the motor 20 may be collected in the oil reservoir 16 provided at the lower portion of the casing 10.

[0071] The rotor 22 may be rotatably disposed at the center of the stator 21. The rotor 22 may be disposed so as to maintain a certain gap with the inner surface of the stator 21.

[0072] A shaft hole 29 may be provided at the center of the rotor 22 that penetrates the rotor 22 in the longitudinal direction. A plurality of refrigerant holes 27 may be provided around the shaft hole 29 of the rotor 22. The plurality of refrigerant holes 27 may be formed to penetrate the rotor 22 in the longitudinal direction, that is, in the up-down direction.

[0073] The refrigerant discharged from the compression part 40 below the motor 20 may move to the upper side of the motor 20 through the gap between the rotor 22 and the stator 21 and the plurality of refrigerant holes 27.

[0074] A drive shaft 30 may be inserted into and fixed in the shaft hole 29 penetrating the center of the rotor 22. Therefore, when power is applied to the motor 20, the rotor 22 may rotate by the electromagnetic force acting between the stator 21 and the rotor 22, and the drive shaft 30 may rotate integrally with the rotor 22.

[0075] When the drive shaft 30 rotates by the motor 20, the compression part 40 may operate to compress the refrigerant.

[0076] The drive shaft 30 may be formed to extend downward from the motor 20. The lower portion of the drive shaft 30 extending downward from the motor 20 may be connected to the compression part 40. The lower portion of the drive shaft 30 may be configured as a crank shaft to operate the compression part 40.

[0077] The crank shaft of the drive shaft 30 may include two eccentric portions, that is, an upper eccentric portion 31 and a lower eccentric portion 32. Therefore, when the drive shaft 30 rotates, the upper eccentric portion 31 and the lower eccentric portion 32 of the crank shaft may rotate integrally with the drive shaft 30.

[0078] The upper eccentric portion 31 may be formed in a cylindrical shape having a diameter larger than the diameter of the drive shaft 30. The center line of the upper eccentric portion 31 may be eccentric with the center line of the drive shaft 30. An upper roller 33 may be disposed on the outer circumferential surface of the upper eccentric portion 31.

[0079] The lower eccentric portion 32 may be disposed below the upper eccentric portion 31, and may be formed in the same manner as the upper eccentric portion 31. In other words, the lower eccentric portion 32 may be formed in a cylindrical shape having a diameter larger than the diameter of the drive shaft 30. The center line of the lower eccentric portion 32 may be eccentric with the center line of the drive shaft 30. A lower roller 34 may be disposed on the outer circumferential surface of the lower eccentric portion 32.

[0080] The lower eccentric portion 32 may be formed to be eccentric in a different direction from the upper eccentric portion 31 with respect to the center line of the drive shaft 30. For example, the lower eccentric portion 32 may be eccentric 180 degrees opposite to the upper eccentric portion 31 with respect to the center line of the drive shaft 30.

[0081] The drive shaft 30 may be rotatably supported by an upper flange 91 and a lower flange 92. The upper flange 91 may be disposed to support the drive shaft 30 between the rotor 22 and the upper eccentric portion 31. The lower flange 92 may be disposed to support the lower end portion of the drive shaft 30 below the lower eccentric portion 32.

[0082] The compression part 40 may be disposed below the motor 20. The compression part 40 may be configured to compress the refrigerant and discharge the compressed refrigerant to the upper side of the compression part 40 according to the rotation of the drive shaft 30.

[0083] Hereinafter, the compression part 40 of the rotary compressor 1 according to an embodiment of the disclosure will be described in detail with reference to FIGS. 4 to 7.

[0084] FIG. 4 is a perspective view illustrating a compression part 40 of a rotary compressor 1 according to an embodiment of the disclosure. FIG. 5 is a bottom view of the compression part 40 of the rotary compressor 1 of FIG. 4. FIG. 6 is a cross-sectional view illustrating the compression part 40 of the rotary compressor 1 of FIG. 4. FIG. 7 is an exploded perspective view illustrating the compression part 40 of the rotary compressor 1 of FIG. 4.

[0085] The compression part 40 may be disposed at the lower portion of the casing 10, and may be configured to suck, compress, and discharge refrigerant by operating by the drive shaft 30 that is rotated by the motor 20.

[0086] Referring to FIGS. 4 to 7, the compression part 40 may be disposed between the upper flange 91 and the lower flange 92 that rotatably support the drive shaft 30.

[0087] The compression part 40 may include an upper compression part 41, a lower compression part 42, and an intermediate plate 70 provided between the upper compression part 41 and the lower compression part 42.

[0088] The upper compression part 41 may be configured to suck and compress the refrigerant according to the rotation of the drive shaft 30. The lower compression part 42 may be provided below the upper compression part 41 and may be configured to suck and compress the refrigerant according to the rotation of the drive shaft 30.

[0089] The upper compression part 41 may be disposed on the upper surface of the intermediate plate 70 and may include an upper cylinder 50 having a flat plate shape. The upper cylinder 50 may include a compression chamber. The compression chamber may be formed as a hollow 51 having a circular cross-section.

[0090] The upper roller 33 disposed in the upper eccentric portion 31 of the drive shaft 30 may be accommodated and rotated in the hollow 51 of the upper cylinder 50.

[0091] The upper compression part 41 may include a refrigerant inlet passage 52 connected to the refrigerant inlet 13 provided in the casing 10. The refrigerant inlet passage 52 may be formed in the upper cylinder 50.

[0092] The refrigerant inlet passage 52 may be formed as a through hole that connects the hollow 51 and the outer circumferential surface of the upper cylinder 50. Therefore, the refrigerant may be introduced into the hollow 51 of the upper cylinder 50 through the refrigerant inlet 13 and the refrigerant inlet passage 52.

[0093] The upper compression part 41 may include an upper discharge port 501 through which the compressed refrigerant is discharged. The upper discharge port 501 may be provided on the upper surface of the upper cylinder 50.

[0094] When the upper roller 33 rotates by the drive shaft 30, the refrigerant may be introduced into the hollow 51 of the upper cylinder 50 through the refrigerant inlet passage 52, compressed by the upper roller 33, and discharged through the upper discharge port 501.

[0095] The lower compression part 42 may be disposed on the lower surface of the intermediate plate 70 and may have a lower cylinder 60 having a flat plate shape. The lower

cylinder 60 may include a compression chamber. The compression chamber may be formed as a hollow 61 having a circular cross-section.

[0096] The lower roller 34 disposed in the lower eccentric portion 32 of the drive shaft 30 may be accommodated and rotated in the hollow 61 of the lower cylinder 60.

[0097] The lower compression part 42 may include a refrigerant inlet passage 62 connected to the refrigerant inlet 13 provided in the casing 10. The refrigerant inlet passage 62 may be formed in the lower cylinder 60.

[0098] The refrigerant inlet passage 62 may be formed as a through hole that connects the hollow 61 and the outer circumferential surface of the lower cylinder 60. Therefore, the refrigerant may be introduced into the hollow 61 of the lower cylinder 60 through the refrigerant inlet 13 and the refrigerant inlet passage 62.

[0099] The lower compression part 42 may include a lower discharge port through which the compressed refrigerant is discharged. The lower discharge port may be provided on the lower surface of the lower cylinder 60. Therefore, the refrigerant compressed by the lower compression part 42 may be discharged below the lower compression part 42 through the lower discharge port.

[0100] When the lower roller 34 rotates by the drive shaft 30, the refrigerant may be introduced into the hollow 61 of the lower cylinder 60 through the refrigerant inlet passage 62, compressed by the lower roller 34, and discharged through the lower discharge port.

[0101] The low-pressure refrigerant may be supplied to the upper compression part 41 and the lower compression part 42 through the accumulator 3.

[0102] The intermediate plate 70 may be disposed between the upper cylinder 50 and the lower cylinder 60. The intermediate plate 70 may be formed in a flat plate shape. Accordingly, the lower cylinder 60, the intermediate plate 70, and the upper cylinder 50 may be laminated to form the compression part 40. The lower cylinder 60, the intermediate plate 70, and the upper cylinder 50 may be integrated connected by a plurality of bolts 93.

[0103] The upper flange 91 may be disposed on the upper surface of the upper cylinder 50. The upper flange 91 may be fixed to the inner circumferential surface of the casing 10. Therefore, when the upper cylinder 50 is fixed to the upper flange 91, the upper cylinder 50 may be fixed to the casing 10.

[0104] The upper flange 91 may be configured to rotatably support the drive shaft 30 and block the upper end of the hollow 51 of the upper cylinder 50.

[0105] The upper flange 91 may be provided with an upper through-hole communicating with the upper discharge port 501 of the upper cylinder 50. Accordingly, the refrigerant discharged through the upper discharge port 501 of the upper cylinder 50 may be discharged to the upper side of the upper flange 91 through the upper through-hole of the upper flange 91.

[0106] The upper flange 91 may include an upper discharge valve 911 configured to open and close the upper through-hole. Therefore, the upper through-hole of the upper flange 91 may be opened and closed by the upper discharge valve 911. When the refrigerant introduced into the upper cylinder 50 is compressed, the upper discharge valve 911 may be opened so that the refrigerant may be discharged to the upper side of the upper flange 91.

[0107] An upper muffler 71 may be disposed on the upper side of the upper flange 91. The upper muffler 71 may be configured to reduce noise generated by the refrigerant discharged through the upper through-hole of the upper flange 91.

[0108] The upper flange 91 may include a plurality of bolt holes 912 provided along the circumferential direction of the upper flange 91.

[0109] The upper flange 91 may include a plurality of upper refrigerant holes 913 provided along the circumferential direction of the upper flange 91. The refrigerant discharged from the lower cylinder 60 may flow to the upper side of the upper flange 91 through the plurality of upper refrigerant holes 913.

[0110] The upper muffler 71 may include a plurality of refrigerant openings 711 through which the refrigerant may pass. The refrigerant passing through the upper flange 91 may be discharged into the space between the motor 20 and the compression part 40 through the plurality of refrigerant openings 711 of the upper muffler 71.

[0111] The upper muffler 71 may be provided with a plurality of bolt holes 712 corresponding to the plurality of bolt holes 912 of the upper flange 91 along the edge of the upper muffler 71.

[0112] A plurality of openings 95 may be provided at the edge of the upper flange 91. The plurality of openings 95 may be formed around the upper muffler 71 disposed on the upper flange 91. The plurality of openings 95 may be formed to penetrate the upper flange 91 upwardly and downwardly. Oil may move to the oil reservoir 16 at the lower portion of the casing 10 through the plurality of openings 95.

[0113] The upper flange 91 may include an upper flange part 915, an upper boss 916, and an upper bearing 917.

[0114] The upper flange part 915 may be formed in a disk shape. The upper flange part 915 may be formed to cover the hollow 51 of the upper cylinder 50. The upper flange part 915 may be formed to have a diameter corresponding to the inner circumferential surface of the casing 10. Therefore, the upper flange 91 may be fixed to the inner circumferential surface of the casing 10.

[0115] The upper flange part 915 may include the upper through-hole communicating with the upper discharge port 501 of the upper cylinder 50. Therefore, the refrigerant discharged through the upper discharge port 501 of the upper cylinder 50 may be discharged to the upper side of the upper flange part 915 through the upper through-hole of the upper flange part 915.

[0116] The upper discharge valve 911 may be provided in the upper through-hole of the upper flange part 915. Accordingly, the upper through-hole of the upper flange part 915 may be opened and closed by the upper discharge valve 911. When the refrigerant introduced into the upper cylinder 50 is compressed above a certain pressure, the upper discharge valve 911 may be opened so that the refrigerant may be discharged to the upper side of the upper flange part 915.

[0117] The upper boss 916 may be vertically extended from the center of the upper flange part 915. The upper boss 916 may be vertically extended upward from the upper flange part 915. A through hole may be formed in the center of the upper boss 916.

[0118] The upper bearing 917 may be disposed in the through hole of the upper boss 916 and may rotatably support the drive shaft 30. Any type of bearing may be used as the upper bearing 917 as long as it can rotatably support

the drive shaft 30. In the case of this disclosure, a sliding bearing may be used as the upper bearing 917.

[0119] The plurality of bolt holes 912 and the plurality of refrigerant holes 913 may be provided in the upper flange part 915 around the upper boss 916. The plurality of openings 95 may be provided on the edge of the upper flange part 915 outside the plurality of bolt holes 912.

[0120] The upper cylinder 50 may be provided with a plurality of tap holes 502 corresponding to the plurality of bolt holes 912 of the upper flange part 915. In addition, the upper cylinder 50 may include a plurality of refrigerant holes 503 corresponding to the plurality of upper refrigerant holes 913 of the upper flange 91.

[0121] When the plurality of bolts 94 are fastened to the plurality of tap holes 502 of the upper cylinder 50, the upper muffler 71 and the upper flange 91 may be fixed to the upper cylinder 50 by the plurality of bolts 94.

[0122] The noise of the refrigerant discharged through the upper through hole of the upper flange 91 may be reduced as the refrigerant passes through the interior of the upper muffler 71, and the refrigerant may be discharged to the upper side of the upper muffler 71, that is, to the space between the motor 20 and the compression part 40 through the plurality of refrigerant openings 711 of the upper muffler 71.

[0123] The lower flange 92 may be disposed on the lower surface of the lower cylinder 60. The lower flange 92 may be configured to rotatably support the lower end portion of the drive shaft 30 and block the lower end of the hollow 61 of the lower cylinder 60.

[0124] The lower flange 92 may be provided with a lower through-hole 924 communicating with the lower discharge port of the lower cylinder 60. Accordingly, the refrigerant discharged through the lower discharge port of the lower cylinder 60 may be discharged to the lower side of the lower flange 92 through the lower through-hole 924 of the lower flange 92.

[0125] The lower flange 92 may include a lower discharge valve 921 configured to open and close the lower through hole 924. Therefore, the lower through-hole 924 of the lower flange 92 may be opened and closed by the lower discharge valve 921. When the refrigerant introduced into the lower cylinder 60 is compressed to a certain pressure or higher, the lower discharge valve 921 may be opened so that the refrigerant may be discharged to the lower side of the lower flange 92.

[0126] A flat muffler 72 may be disposed on the lower side of the lower flange 92. The flat muffler 72 may be configured to reduce noise generated by the refrigerant discharged through the lower through-hole 924 of the lower flange 92.

[0127] In addition, the lower flange 92 may be configured so that the refrigerant discharged through the lower through-hole 924 is not discharged below the flat muffler 72.

[0128] The lower flange 92 may include a plurality of first bolt holes 922 provided along the circumferential direction of the lower flange 92.

[0129] The lower flange 92 may include a plurality of lower refrigerant holes 923 provided along the circumferential direction of the lower flange 92. The refrigerant discharged from the lower through-hole 924 of the lower flange 92 may flow to the upper side of the lower flange 92 through the plurality of lower refrigerant holes 923.

[0130] In detail, the plurality of lower refrigerant holes 923 of the lower flange 92 may be formed to coincide with

the plurality of refrigerant holes 603 of the lower cylinder 60, the plurality of refrigerant holes 703 of the intermediate plate 70, the plurality of refrigerant holes 503 of the upper cylinder 50, and the plurality of upper refrigerant holes 913 of the upper flange 91.

[0131] Accordingly, the refrigerant discharged from the lower through hole 924 of the lower flange 92 may move to the space formed by the upper flange 91 and the upper muffler 71 through the plurality of lower refrigerant holes 923 of the lower flange 92, the plurality of refrigerant holes 603 of the lower cylinder 60, the plurality of refrigerant holes 703 of the intermediate plate 70, the plurality of refrigerant holes 503 of the upper cylinder 50, and the plurality of upper refrigerant holes 913 of the upper flange 91.

[0132] The refrigerant in the space between the upper flange 91 and the upper muffler 71 may move to the upper side of the upper muffler 71 through the plurality of openings 711 of the upper muffler 71.

[0133] The flat muffler 72 may not include an opening or hole through which the refrigerant may be discharged. Therefore, the refrigerant discharged through the lower through hole 924 of the lower flange 92 may not be discharged into the oil in the oil reservoir 16 in which the flat muffler 72 is submerged through the flat muffler 72.

[0134] The lower flange 92 may include a flange part 925, a boss 926, and a bearing 927.

[0135] The flange part 925 may be formed in a disk shape. The flange part 925 may be formed to cover the hollow 61 of the lower cylinder 60. The flange part 925 may be formed in a size corresponding to the lower cylinder 60.

[0136] The flange part 925 may include the lower through-hole 924 communicating with the lower discharge port of the lower cylinder 60. Therefore, the refrigerant discharged through the lower discharge port of the lower cylinder 60 may be discharged below the flange part 925 through the lower through-hole 924 of the flange part 925.

[0137] The lower discharge valve 921 may be provided in the lower through-hole 924 of the flange part 925. Therefore, the lower through-hole 924 of the flange part 925 may be opened and closed by the lower discharge valve 921. When the refrigerant introduced into the lower cylinder 60 is compressed above a certain pressure, the lower discharge valve 921 may be opened so that the refrigerant may be discharged below the flange part 925.

[0138] The boss 926 may extend vertically from the center of the flange part 925. The boss 926 may extend vertically upward from the flange part 925. A through hole may be formed in the center of the boss 926.

[0139] The bearing 927 may be disposed in the through hole of the boss 926 and may rotatably support the drive shaft 30. Accordingly, the drive shaft 30 may be rotatably supported by the bearing 927 and the upper bearing 917. Any type of bearing may be used as the bearing 927 as long as it can rotatably support the drive shaft 30. In the case of this disclosure, a sliding bearing may be used as the bearing 927.

[0140] The plurality of first bolt holes 922 and the plurality of lower refrigerant holes 923 may be provided in the flange part 925 around the boss 926.

[0141] The lower cylinder 60 may be provided with a plurality of bolt holes 602 corresponding to the plurality of first bolt holes 922 of the flange part 925. In addition, the lower cylinder 60 may include the plurality of refrigerant

holes 603 corresponding to the plurality of lower refrigerant holes 923 of the lower flange 92.

[0142] The flat muffler 72 may be disposed on the lower surface of the flange part 925 of the lower flange 92. Hereinafter, the flat muffler 72 will be described in detail with reference to FIGS. 8 and 9.

[0143] FIG. 8 is a perspective view illustrating a flat muffler 72 used in a rotary compressor 1 according to an embodiment of the disclosure. FIG. 9 is a perspective view illustrating the flat muffler 72 of FIG. 8 as seen from above.

[0144] Referring to FIGS. 8 and 9, the flat muffler 72 may be formed in an approximately dome shape and may include a plurality of concave portions 721 formed at regular intervals along the circumferential direction of the flat muffler 72 on the side surface of the flat muffler 72. In detail, the flat muffler 72 may have a disk shape and the central portion of the disk may be formed to protrude downward in an approximately dome shape.

[0145] In addition, the flat muffler 72 may include a plurality of second bolt holes 722 formed at positions corresponding to the plurality of concave portions 721 along the edge of the flat muffler 72. The plurality of second bolt holes 722 may be formed to correspond to the plurality of first bolt holes 922 of the lower flange 92.

[0146] For example, the flat muffler 72 may include a fixing plate 723, a caulking portion 724, and a muffler portion 725.

[0147] The fixing plate 723 may be formed in an approximately ring shape. The fixing plate 723 may include an outer circumferential surface formed in a circular shape and an inner circumferential surface concentric with the outer circumferential surface and having a diameter smaller than that of the outer circumferential surface. The inner circumferential surface of the fixing plate 723 may include a plurality of protrusions 7231. The plurality of protrusions 7231 of the inner circumferential surface may be formed to protrude toward the center of the fixing plate 723. The plurality of protrusions 7231 may be formed to correspond to the plurality of concave portions 721 of the muffler portion 725. The plurality of protrusions 7231 may be formed in a curve corresponding to the plurality of concave portions 721 of the muffler portion 725, respectively.

[0148] The portion between two adjacent protrusions 7231 may be concave to form a recess 7232. In other words, the plurality of protrusions 7231 may form a plurality of recesses 7232. Because each of the plurality of protrusions 7231 forms a projecting portion, the plurality of recesses 7232 and the plurality of projecting portions may be alternately provided on the inner circumferential surface of the fixing plate 723. Accordingly, the inner circumferential surface of the fixing plate 723 may be formed concavely and convexly by the plurality of protrusions 7231.

[0149] The fixing plate 723 may be fixed to the flange part 925 of the lower flange 92. The fixing plate 723 may include the plurality of second bolt holes 722. The plurality of second bolt holes 722 may be formed in the plurality of protrusions 7231 of the fixing plate 723, respectively. The plurality of second bolt holes 722 may be formed to correspond to the plurality of first bolt holes 922 of the lower flange 92.

[0150] The muffler portion 725 may be formed by extending downward from the inner circumferential surface of the fixing plate 723. The muffler portion 725 may be formed to protrude in an approximately dome shape from the lower

surface of the fixing plate 723. The plurality of concave portions 721 may be formed at regular intervals in the circumferential direction on the side surface of the muffler portion 725. The plurality of concave portions 721 may be formed as curved surfaces that protrude into the interior of the muffler portion 725.

[0151] As illustrated in FIG. 9, the plurality of concave portions 721 may protrude into the interior space of the muffler portion 725. Therefore, the interior space of the muffler portion 725 may be formed concavely and convexly.

[0152] A through hole 726 may be provided at the lower end of the muffler portion 725. An oil pump disposed at the lower end of the drive shaft 30 may be immersed in the oil reservoir 16 through the through hole 726 of the muffler portion 725.

[0153] The caulking portion 724 may be provided at the lower end of the muffler portion 725. The caulking portion 724 may be provided at the edge of the through hole 726. The caulking portion 724 may be formed to be fixed to one end of the boss 926 of the lower flange 92. When the caulking portion 724 of the muffler portion 725 is joined to one end of the boss 926 of the lower flange 92, the refrigerant may be prevented from leaking between one end of the boss 926 of the lower flange 92 and the through hole 726 of the muffler portion 725.

[0154] The caulking portion 724 may be formed by performing a caulking operation along the entire circumference of the through hole 726 of the muffler portion 725.

[0155] The muffler portion 725 may be provided between the fixing plate 723 and the caulking portion 724.

[0156] The upper end of the flat muffler 72 may be formed not to protrude above the lower surface of the lower flange 92. In detail, the fixing plate 723 of the flat muffler 72 may not protrude above the lower surface of the flange part 925 of the lower flange 92, but may be positioned below the lower surface of the flange part 925. In other words, the upper surface of the fixing plate 723 of the flat muffler 72 may be in contact with the lower surface of the flange part 925 of the lower flange 92.

[0157] The upper surface of the fixing plate 723 of the flat muffler 72 may be formed not to be in contact with the side surface of the flange part 925 of the lower flange 92. When the fixing plate 723 of the flat muffler 72 does not contact the side surface of the flange part 925 of the lower flange 92, pressure may not be applied to the bearing 927 disposed in the lower flange 92 by the flat muffler 72.

[0158] The sealing plate 100 may be disposed between the lower flange 92 and the flat muffler 72. The sealing plate 100 may be formed to prevent or minimize the leakage of refrigerant between the lower flange 92 and the flat muffler 72.

[0159] The sealing plate 100 may be positioned between the lower surface of the flange part 925 of the lower flange 92 and the upper surface of the fixing plate 723 of the flat muffler 72.

[0160] FIG. 10 is a perspective view illustrating a sealing plate 100 used in a rotary compressor 1 according to an embodiment of the disclosure.

[0161] Referring to FIG. 10, the sealing plate 100 may be formed in a flat plate having an approximately ring shape. The sealing plate 100 may be formed to correspond to the fixing plate 723 of the flat muffler 72. The sealing plate 100 may be formed to seal the flat portion between the flat muffler 72 and the lower flange 92.

[0162] The sealing plate 100 may include an outer circumferential surface formed in a circular shape and an inner circumferential surface concentric with the outer circumferential surface and having a diameter smaller than that of the outer circumferential surface.

[0163] The inner circumferential surface of the sealing plate 100 may include a plurality of bolt seats 101 and a plurality of refrigerant grooves 103.

[0164] The plurality of bolt seats 101 may be formed to protrude from the inner circumferential surface of the sealing plate 100 toward the center of the sealing plate 100. The plurality of bolt seats 101 may be formed to correspond to the plurality of protrusions 7231 of the fixing plate 723. Accordingly, the plurality of bolt seats 101 may correspond to the plurality of concave portions 721 of the muffler portion 725. The plurality of bolt seats 101 may be formed in a curve corresponding to the plurality of concave portions 721 of the flat muffler 72, respectively.

[0165] A plurality of third bolt holes 102 corresponding to the plurality of second bolt holes 722 of the flat muffler 72 may be formed in the plurality of bolt seats 101, respectively. In addition, the plurality of third bolt holes 102 of the sealing plate 100 may be formed to correspond to the plurality of first bolt holes 922 of the lower flange 92.

[0166] The plurality of refrigerant grooves 103 may be formed so that the sealing plate 100 does not cover the plurality of lower refrigerant holes 923 of the lower flange 92. Therefore, when the sealing plate 100 is disposed on the lower surface of the lower flange 92, the sealing plate 100 may not cover the plurality of lower refrigerant holes 923.

[0167] The plurality of refrigerant grooves 103 may be formed between the plurality of bolt seats 101. In other words, the plurality of refrigerant grooves 103 and the plurality of bolt seats 101 may be formed alternately in the circumferential direction of the sealing plate 100.

[0168] Each of the plurality of bolt seats 101 may form a convex portion, and each of the plurality of refrigerant grooves 103 may form a concave portion. Accordingly, the inner circumferential surface of the sealing plate 100 may include a plurality of convex portions and a plurality of concave portions. In other words, the inner circumferential surface of the sealing plate 100 may be formed concavely and convexly. Accordingly, the inner circumferential surface of the sealing plate 100 may be formed concavely and convexly corresponding to the flat muffler 72.

[0169] The sealing plate 100 may be formed of a material capable of preventing refrigerant from leaking between the flat muffler 72 and the lower flange 92. For example, the sealing plate 100 may be formed of a heat-resistant resin, steel, copper, or the like. In addition, the sealing plate 100 may be formed of a material in which at least two of the heat-resistant resin, steel, and copper are laminated in a plate shape.

[0170] The flat muffler 72, the sealing plate 100, and the lower flange 92 may be fixed to the upper cylinder 50 by the plurality of bolts 93. In detail, the plurality of bolts 93 may be fastened to the plurality of tap holes 502 of the upper cylinder 50 through the plurality of second bolt holes 722 of the flat muffler 72, the plurality of third bolt holes 102 of the sealing plate 100, the plurality of first bolt holes 922 of the lower flange 92, the plurality of bolt holes 602 of the lower cylinder 60, and the plurality of bolt holes 702 of the intermediate plate 70. Then, the flat muffler 72, the sealing

plate 100, the lower flange 92, the lower cylinder 60, and the intermediate plate 70 may be integrally fixed to the upper cylinder 50.

[0171] The refrigerant discharged downward through the lower through hole 924 of the lower flange 92 may reduce in noise while passing through the inside of the flat muffler 72, and then flow into the plurality of lower refrigerant holes 923 of the lower flange 92.

[0172] The refrigerant introduced into the plurality of lower refrigerant holes 923 of the lower flange 92 may be discharged to the upper side of the upper flange 91 through the plurality of refrigerant holes 603 of the lower cylinder 60, the plurality of refrigerant holes 703 of the intermediate plate 70, the plurality of refrigerant holes 503 of the upper cylinder 50, and the plurality of upper refrigerant holes 913 of the upper flange 91.

[0173] The refrigerant discharged to the upper side of the upper flange 91 may be discharged to the upper side of the upper muffler 71, that is, the space between the motor 20 and the compression part 40, through the plurality of openings 711 of the upper muffler 71.

[0174] The refrigerant that has moved to the space between the motor 20 and the compression part 40 may move to the upper side of the motor 20 through the motor 20. For example, the refrigerant in the space between the motor 20 and the compression part 40 may move to the upper side of the motor 20 through the gap between the rotor 22 and the stator 21 and the plurality of refrigerant holes 27 provided in the rotor 22.

[0175] The refrigerant that has moved to the upper side of the motor 20 may be discharged to the outside of the casing 10 through the refrigerant discharge pipe 14 disposed in the upper casing 11.

[0176] Hereinafter, the positional relationship of the lower flange 92, the flat muffler 72, and the sealing plate 100 will be described in detail with reference to FIGS. 11 to 13.

[0177] FIG. 11 is a view illustrating a state in which a sealing plate 100 and a flat muffler 72 are disposed in a lower flange 92. FIG. 12 is a bottom view of FIG. 11. FIG. 13 is a bottom view illustrating a state in which the flat muffler 72 is removed from FIG. 11.

[0178] Referring to FIGS. 11 to 13, the flat muffler 72 may be disposed on the lower surface of the lower flange 92. The sealing plate 100 may be disposed between the lower flange 92 and the flat muffler 72.

[0179] The sealing plate 100 may be disposed on the lower surface of the lower flange 92 so that the plurality of third bolt holes 102 thereof are aligned with the plurality of first bolt holes 922 of the lower flange 92.

[0180] When the plurality of third bolt holes 102 of the sealing plate 100 are aligned with plurality of first bolt holes 922 of the lower flange 92, the plurality of refrigerant grooves 103 of the sealing plate 100 may be positioned at the edges of the plurality of lower refrigerant holes 923 of the lower flange 92.

[0181] At this time, the minimum distance between the inner circumferential surface of the sealing plate 100 and the plurality of lower refrigerant holes 923 of the lower flange 92 may be zero (0). For example, as illustrated in FIG. 13, when the inner surface of the lower refrigerant hole 923 is in contact with the bottom surface of the refrigerant groove 103 of the sealing plate 100, the distance between the lower refrigerant hole 923 and the inner circumferential surface of the sealing plate 100 may be zero (0).

[0182] Alternatively, the inner circumferential surface of the sealing plate 100 may be formed so that the bottom surfaces of the refrigerant grooves 103 of the sealing plate 100 are spaced apart from the inner surfaces of the lower refrigerant holes 923. In this case, the distance between the lower refrigerant hole 923 and the inner circumferential surface of the sealing plate 100 may be greater than zero (0). At this time, the distance between the lower refrigerant hole 923 and the inner circumferential surface of the sealing plate 100 may be defined to be smaller than the minimum distance between the outer circumferential surface of the lower flange 92 and the lower refrigerant hole 923. When the distance between the lower refrigerant hole 923 and the inner circumferential surface of the sealing plate 100 is greater than the minimum distance between the outer circumferential surface of the lower flange 92 and the lower refrigerant hole 923, the refrigerant may leak.

[0183] In addition, in order to prevent refrigerant leakage, the distance between the bottom surface of the refrigerant groove 103 of the sealing plate 100 and the outer circumferential surface of the sealing plate 100 may be at least 1 mm. In other words, the distance between the concave portion of the inner circumferential surface of the sealing plate 100 and the outer circumferential surface of the sealing plate 100 may be at least 1 mm.

[0184] When the sealing plate 100 is formed with the structure as described above, the plurality of lower refrigerant holes 923 of the lower flange 92 may be exposed through the plurality of refrigerant grooves 103 of the sealing plate 100. In other words, the plurality of lower refrigerant holes 923 of the lower flange 92 may not be covered by the sealing plate 100.

[0185] Therefore, the refrigerant in the space between the lower flange 92 and the flat muffler 72 may flow into the lower refrigerant holes 923 of the lower flange 92 without interference of the sealing plate 100.

[0186] In the case where the flat muffler 72 is disposed on the lower surface of the lower flange 92, when the space between the lower flange 92 and the flat muffler 72 is filled with compressed refrigerant, as illustrated in FIG. 14, a portion of the flat muffler 72 may be opened so that the refrigerant may leak therethrough.

[0187] FIG. 14 is a view illustrating a state in which a portion of a flat muffler 72 is opened by compressed refrigerant. For reference, in FIG. 14, the opened portion is exaggerated to show the opened state of the flat muffler 72.

[0188] Referring to FIG. 14, reference numeral G indicates the opening of the flat muffler 72. In other words, the opening of the flat muffler 72 refers to the vertical distance between the protrusion 7231 and the concave portion of the fixing plate 723.

[0189] When the rotary compressor 1 operates so that refrigerant is discharged into the space between the lower flange 92 and the flat muffler 72, portions of the fixing plate 723 between the plurality of protrusions 7231 of the fixing plate 723 at which the plurality of bolts 93 that fix the flat muffler 72 to the lower flange 92 are disposed may be deformed by the pressure of the compressed refrigerant. In other words, the concave portion of the fixing plate 723 may be deformed downward from the protrusion 7231 thereof by the pressure of the refrigerant.

[0190] When there is no compressed refrigerant between the lower flange 92 and the flat muffler 72, the plurality of concave portions of the fixing plate 723 of the flat muffler 72 may be restored to their original state and become flat with the plurality of protrusions 7231. In other words, the opening G of the flat muffler 72 may be zero (0).

[0191] In order to minimize deformation of the bearing 927 of the lower flange 92 and to minimize the amount of refrigerant leaking between the lower flange 92 and the flat muffler 72, the sealing plate 100 may be formed to have a certain thickness Ts. For example, the thickness Ts of the sealing plate 100 may be formed to have a certain thickness ratio with the thickness Tm of the flat muffler 72.

[0192] The thickness Ts of the sealing plate 100 may be formed to have the following relationship with the thickness Tm of the flat muffler 72:

[0193] $0.5 \leq \text{thickness Ts of the sealing plate 100} / \text{thickness Tm of the flat muffler 72} \leq 2.0$

[0194] FIG. 15 is a graph illustrating a change in deformation of a bearing 927 and opening of a flat muffler 72 according to the thickness of a sealing plate 100.

[0195] FIG. 15 illustrates the results of calculating the deformation of the bearing 927 disposed in the lower flange 92 and the opening G of the flat muffler 72 by computer simulation when the thickness Tm of the flat muffler 72 is 1.2 mm and the thickness Ts of the sealing plate 100 is 0.5 mm, 1 mm, 2 mm, 2.5 mm, and 3 mm. Here, the opening G of the flat muffler 72 refers to the vertical distance between the protrusion 7231 and the concave portion of the fixing plate 723 of the flat muffler 72.

[0196] Referring to FIG. 15, when the thickness Ts of the sealing plate 100 increases, the deformation of the concave portions between the plurality of protrusions 7231 of the fixing plate 723 of the flat muffler 72, that is, the opening G of the flat muffler 72 may be reduced. Accordingly, when the thickness Ts of the sealing plate 100 increases, the leakage of refrigerant between the flat muffler 72 and the lower flange 92 may be minimized or prevented.

[0197] In addition, when the thickness Ts of the sealing plate 100 increases, the deformation of the bearing 927 disposed in the lower flange 92 may increase. In detail, when the thickness Ts of the sealing plate 100 increases, the movement distance of the bolt 93 may increase, which may increase the stress generated in the boss 926 of the lower flange 92. As the stress of the boss 926 increases, the deformation of the bearing 927 disposed in the boss 926 may increase.

[0198] When the deformation of the bearing 927 of the lower flange 92 increases, the input of the rotary compressor 1 may increase. When the input of the rotary compressor 1 increases, the energy efficiency (EER: energy efficiency ratio) may decrease.

[0199] When the deformation of the bearing 927 decreases, there is room for maintaining the oil film thickness, so that a design that reduces the diameter of the bearing is possible and the reliability of the rotary compressor 1 may be improved.

[0200] When a skirt type muffler is used, the bearing deformation is 10.5 μm . When the flat muffler 72 according to an embodiment of the disclosure and the sealing plate 100 having a thickness of 1 mm are used, the bearing deformation may be reduced to 6.7 μm .

[0201] Therefore, the thickness Ts of the sealing plate 100 may be defined so as to minimize the bearing deformation and the opening G of the flat muffler 72.

[0202] Referring to FIG. 15 again, in the case where the thickness Tm of the flat muffler 72 is 1 mm, when the thickness Ts of the sealing plate 100 is 0.5 mm or less, the bearing deformation is 6.7 μm , which may not increase the input of the rotary compressor 1, but the opening of the flat muffler 72 becomes 18.0 μm , which may cause excessive leakage of the refrigerant.

[0203] When the thickness Ts of the sealing plate 100 is 2.5 mm or more, the opening G of the flat muffler 72 becomes 12.0 μm , so that the leakage of the refrigerant may be appropriate, but the bearing deformation is 7.3 μm , which may increase the input of the rotary compressor 1. Therefore, the thickness Ts of the sealing plate 100 may be more than 0.5 mm and less than 2.5 mm.

[0204] For example, considering the ratio of the thickness Ts of the sealing plate 100 to the thickness Tm of the flat muffler 72, when the thickness Tm of the flat muffler 72 is 1.2 mm, the thickness Ts of the sealing plate 100 may be 0.6 mm to 2.4 mm.

[0205] The performance of the rotary compressor 1 according to an embodiment of the disclosure having the above structure was compared with the performance of a rotary compressor according to the prior art using a skirt muffler.

[0206] Table 1 below is a performance comparison table of the rotary compressor 1 according to an embodiment of the disclosure and the rotary compressor according to the prior art using a skirt muffler. FIG. 16 is a graph illustrating a change rate in energy efficiency of a rotary compressor 1 according to an embodiment of the disclosure relative to the energy efficiency of a rotary compressor according to the prior art using a skirt muffler.

[0207] The comparative tests below were performed under ASHRAE-T conditions. ASHRAE-T conditions are condensing temperature of 54.4° C., liquid temp of 46.1° C., evaporating temperature of 7.2° C., suction temperature of 35° C., and ambient temperature of 35° C.

TABLE 1

operating speed	cooling power (Btu/h)		input (W)		EER	
	related art	disclosure	related art	disclosure	related art	disclosure
30	23,906	+0.5%	2,058	-1.2%	11.61	+1.7%
45	36,598	+0.4%	3,079	-0.6%	11.88	+1.1%
60	49,947	+0.4%	4,246	-0.5%	11.76	+0.8%

[0208] In Table 1, the prior art refers to a rotary compressor having a skirt muffler disposed on the lower flange 92. The disclosure refers to the rotary compressor 1 having the flat muffler 72 and the sealing plate 100 disposed on the lower flange 92. Here, the skirt muffler is formed to include a skirt that wraps around the side surface of the flange part 925 of the lower flange 92. In other words, the skirt muffler includes the skirt that extends upward from the outer circumferential surface of the fixing plate 723 of the flat muffler 72 according to an embodiment of the disclosure. The unit of the operating speed is revolutions per second (rps). EER represents energy efficiency. Referring to Table 1, it can be seen that the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure has increased cooling power compared to the rotary compressor according to the prior art. In other words, the cooling power of the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure may increase by 0.4% to 0.5% compared to the cooling power of the rotary compressor according to the prior art. Accordingly, it can be seen that the refrigerant leakage between the flat muffler 72 and the lower flange 92 is minimized by the sealing plate 100.

[0209] In addition, it can be seen that the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure has a reduced input compared to the rotary compressor according to the prior art. In other words, the input of the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure may be reduced by 0.5% to 1.2% compared to the input of the rotary compressor according to the prior art. Accordingly, it can be seen that the bearing deformation of the lower flange 92 due to the sealing plate 100 and the flat muffler 72 is minimized.

[0210] In addition, it can be seen that the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure has increased energy efficiency compared to the rotary compressor according to the prior art. In other words, the EER of the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure may increase by 0.8% to 1.7% compared to the EER of the rotary compressor according to the prior art. Accordingly, the rotary compressor 1 using the flat muffler 72 and the sealing plate 100 according to an embodiment of the disclosure may increase the EER by an average of 1.2% in the operating range of 30 to 60 rps compared to the rotary compressor according to the prior art.

[0211] As described above, in the rotary compressor 1 according to an embodiment of the disclosure, the flat muffler 72 and the sealing plate 100 may be disposed on the lower flange 92, so that the bearing deformation of the lower flange 92 may be minimized and the refrigerant leakage due to the flat muffler 72 may be minimized, thereby improving the energy efficiency.

[0212] In the foregoing, the disclosure has been shown and described with reference to various embodiments. However, it is understood by those skilled in the art that various changes may be made in form and detail without departing from the scope of the disclosure as defined by the appended claims and equivalents thereof.

1. A rotary compressor comprising:
 - a casing;
 - a motor to be disposed inside the casing;
 - a compression part to be disposed below the motor;
 - a lower flange to be disposed below the compression part;

a muffler, having a portion that is flat, to be disposed on a lower surface of the lower flange; and
a sealing plate to be disposed between the muffler and the lower flange.

2. The rotary compressor of claim 1, wherein the portion that is flat is an upper end of the muffler does not protrude above the lower surface of the lower flange.

3. The rotary compressor of claim 1, wherein the muffler is formed in a dome shape and includes a plurality of concave portions formed at regular intervals along a circumferential direction of the muffler on a side surface of the muffler, and wherein the sealing plate is formed in a ring shape and includes an inner circumferential surface formed concavely and convexly corresponding to the muffler.

4. The rotary compressor of claim 3, wherein the lower flange includes a plurality of lower refrigerant holes and a plurality of first bolt holes formed along an edge of the lower flange, and wherein the muffler includes a plurality of second bolt holes formed corresponding to the plurality of first bolt holes of the lower flange at positions corresponding to the plurality of concave portions along an edge of the muffler.

5. The rotary compressor of claim 4, wherein the sealing plate includes a plurality of bolt seats formed to protrude toward a center of the sealing plate from an inner circumferential surface of the sealing plate and having a plurality of third bolt holes corresponding to the plurality of second bolt holes of the muffler.

6. The rotary compressor of claim 5, wherein the sealing plate is formed so as not to cover the plurality of lower refrigerant holes of the lower flange.

7. The rotary compressor of claim 6, wherein a distance between a concave portion of the sealing plate and an outer circumferential surface of the sealing plate is at least 1 mm.

8. The rotary compressor of claim 1, wherein a thickness of the sealing plate has a relationship with a thickness of the muffler as follows:

$0.5 \leq \text{thickness } T_s \text{ of the sealing plate} / \text{thickness } T_m \text{ of the muffler} \leq 2.0$.

9. The rotary compressor of claim 1, wherein oil is received in a lower portion of the casing, and the muffler is immersed in the oil.

10. The rotary compressor of claim 1, wherein the sealing plate is formed of one of heat-resistant resin, steel, copper, and a material in which at least two of the heat-resistant resin, the steel, and the copper are laminated.

11. The rotary compressor of claim 1, wherein the lower flange comprises:

- a flange part formed in a disk shape;
- a boss extending vertically from the flange part and including a through hole; and
- a bearing disposed in the through hole of the boss, and wherein the muffler comprises:
 - a fixing plate formed in a ring shape and fixed to the flange part of the lower flange;
 - a caulking portion fixed to one end of the boss; and
 - a muffler portion provided between the fixing plate and the caulking portion.

12. The rotary compressor of claim **11**, wherein the muffler portion includes a plurality of concave portions formed at regular intervals in a circumferential direction of the muffler.

13. The rotary compressor of claim **12**, wherein the fixing plate includes a plurality of protrusions corresponding to the plurality of concave portions of the muffler portion.

14. The rotary compressor of claim **13**, wherein the flange part of the lower flange includes a plurality of lower refrigerant holes and a plurality of first bolt holes, and

the plurality of protrusions of the fixing plate includes a plurality of second bolt holes corresponding to the plurality of first bolt holes.

15. The rotary compressor of claim **14**, wherein an inner circumferential surface of the sealing plate includes a plurality of bolt seats formed to correspond to the plurality of protrusions of the fixing plate and a plurality of refrigerant grooves formed so as not to cover the plurality of lower refrigerant holes of the flange part.

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